

CMSC6950 — Fall 2025

Homework #5

Due Date: Work must be submitted to GradeScope by 5pm (St. John's time) on Tuesday, November 18.

- (10 points) Consider the graph shown in Figure 1. Discuss the components of the graphic as reviewed in class (data, variable transformations, data frame, data scale, coordinates, graph, guides).

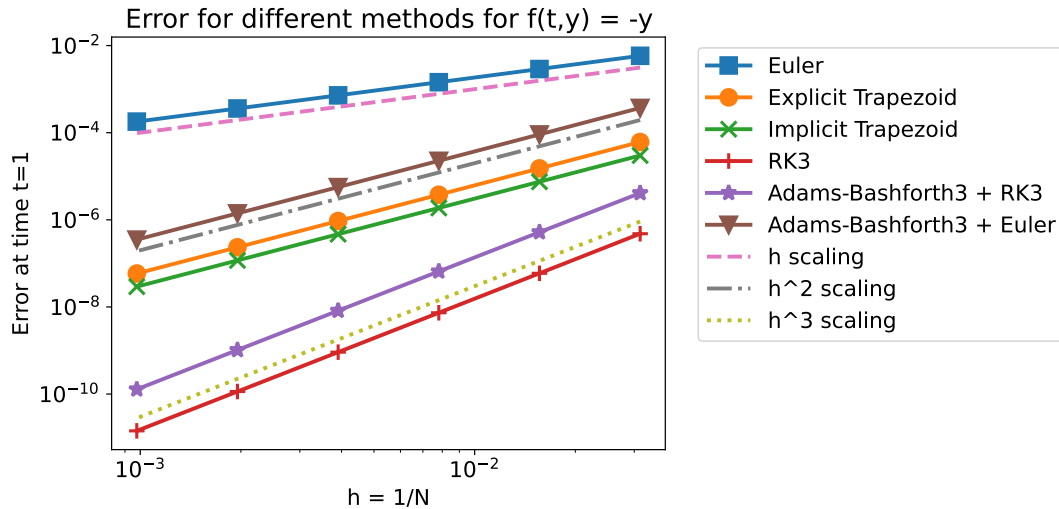


Figure 1: Errors at time $t = 1$ for a variety of numerical methods for solving $y' = f(t, y)$ as a function of time-step size, $1/N$.

- (30 points) Complete the plotting exercises detailed in the `README.md` file from https://github.com/MacLachlanTeaching/CMSC6950_HW5_data. For each data set, be sure to plot it effectively, using the principles and techniques discussed in class. Submit to GradeScope both a listing of your code for each data set as well as the produced graphics.